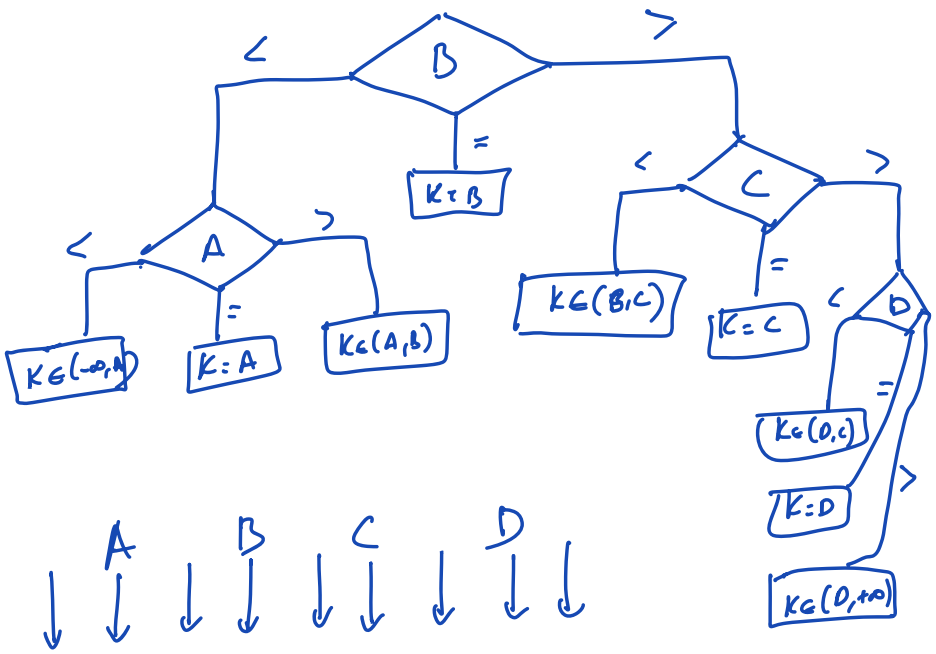
Decision-tree for searching K in a sorted list {A, B, C, D}
(binary search)



9 leaves $\Rightarrow 2n+1$

$h \geq \log_2(9)$ $\quad h \geqslant \log_2(2n+1)$

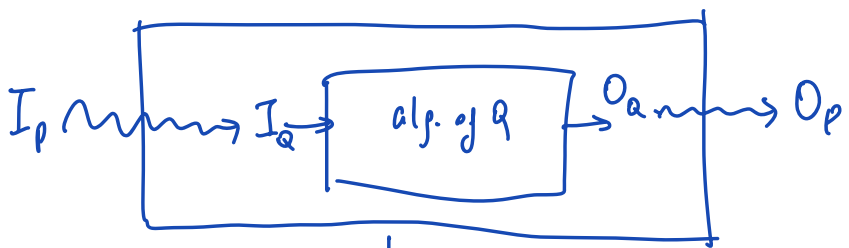


5 leaves $\Rightarrow n+1$

$h \geqslant \log_2(n+1)$

Problem Reduction

$P \longrightarrow Q$



an algorithm of P using the alg. of Q.

Let's say that Q has an alg. with complexity $O(f(n))$

Let's say that the reduction can be done in $O(n)$.

$\Rightarrow$ There is an alg. of P with $O(f(n))$ time complexity.

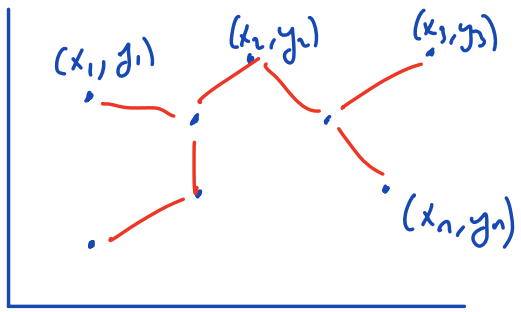
P cannot be more difficult than Q.

Q is at least as hard as P.

if P has a LB of $\Omega(f(n))$, then Q has the same LB.

Element Uniqueness (Problem EU)

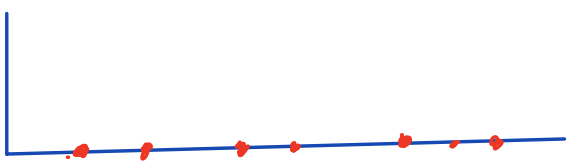
$I_{EU} = \{a_1, a_2, \ldots a_n\}$ $O_{EU} =$ T or F
1 or 0.



$I_{MST} = \{(x_1, y_1), (x_2, y_2) \ldots (x_n, y_n)\}$ $O_{MST} =$ M.S.T.

$I_{EU} \longrightarrow I_{MST}$

$\{a_1, a_2, \ldots a_n\} = \{(a_1, 0), (a_2, 0), \ldots (a_n, 0)\}$



Run MST alg.



In the resulting MST, if there is an edge of length zero, then the elements are not unique. Return False.
If there is no edge with length zero, return True.

Since E.U. problem has a LB of $\Omega(n \log n)$,
MST problem has also the same LB. ($\Omega(n \log n)$)

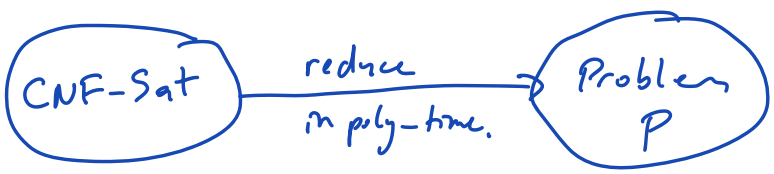
Turing's halting problem:

< "code of an algorithm", "Inputs" >

Output = Yes if the algorithm halts with those "Inputs".
No otherwise.

Decision version of TSP: Is there any Hamiltonian cycle of length $\leq m$?

Generate a potential solution: $a_1\ a_2\ a_3\ a_4 \ldots a_1$
Verify whether this is a solution or not.



Problem P is at least as hard as CNF-Sat.

→ CNF-Sat is at least as hard as any NP problem (Cook's theorem)

Therefore problem P is at least as hard as any NP Problem

If it is also in NP, it is NP-Complete.